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On-line inference for hidden Markov models via particle filters

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Summary. We consider the on-line Bayesian analysis of data by using a hidden Markov model, where inference is tractable conditional on the history of the state of the hidden component. A new particle filter algorithm is introduced and shown to produce promising results when analysing data of this type. The algorithm is similar to the mixture Kalman filter but uses a different resampling algorithm. We prove that this resampling algorithm is computationally efficient and optimal, among unbiased resampling algorithms, in terms of minimizing a squared error loss function. In a practical example, that of estimating break points from well-log data, our new particle filter outperforms two other particle filters, one of which is the mixture Kalman filter, by between one and two orders of magnitude.

Keywords: Changepoints; Ion channel; Kalman filter; Markov chain Monte Carlo methods; Particle filters; Smoothing; Well-log data

1. Introduction

We consider the on-line analysis of hidden Markov models (Rabiner and Juang, 1986). Hidden Markov models are used, for example, in modelling ion channel data (Ball and Rice, 1992), and extensively in bioinformatics (Durbin *et al.*, 1998). They consist of a hidden state variable I_t and a measurable state X_t, which both vary over time. Inference about these two states is based on indirect measurements of X_t.

Standard Markov chain Monte Carlo methods (Green, 1995) are poorly suited to the on-line analysis of evolving data sets. Instead we use sequential Monte Carlo methods, called particle filters (Doucet *et al.*, 2001). We focus on models for which the posterior density of the measurable states $X_1, \ldots, X_t$ is tractable conditional on the history of the hidden states $I_1, \ldots, I_t$. Particle filters are particularly well suited to this type of problem, as the values of the measurable states can be integrated out by using marginalization (Liu *et al.*, 2001), a technique that has previously been used in the mixture Kalman filter of Chen and Liu (2000).

We propose a new filter which is similar to the mixture Kalman filter except for the resampling step. Our new resampling step is optimal, for a class of unbiased resampling algorithms which include most resampling algorithms that are currently used, in terms of minimizing a squared error loss function.

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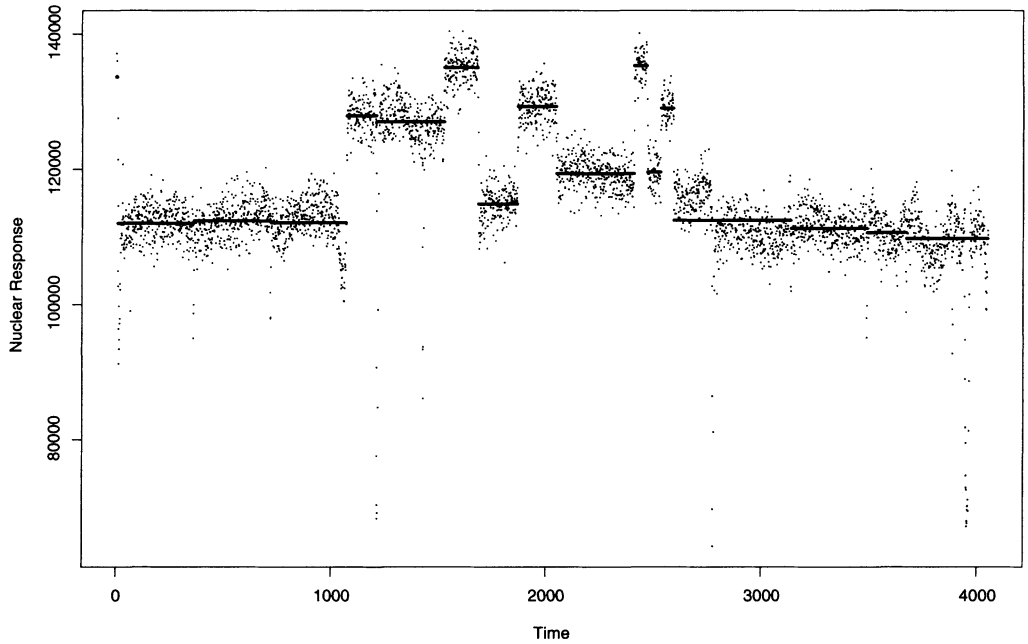

Fig. 1. Well-log data (·) and an estimate of the underlying signal for them (———): the data come from Ó Ruanaidh and Fitzgerald (1996) and originated from Schlumberger

Our work has been motivated by, and will be demonstrated on, the problem of detecting break points in well-log data (Ó Ruanaidh and Fitzgerald, 1996). An example of well-log data is shown in Fig. 1, and consists of measurements of the nuclear magnetic response of underground rocks. The underlying signal is piecewise constant, each constant segment relating to a stratum of a single type of rock. The jump discontinuities in the signal occur each time that a new rock stratum is met.

In drilling for oil, it is important to predict changes in rock type to prevent blow-outs: sudden uncontrolled flows of drilling fluid, oil or water, up the borehole, which are caused when the pressure exerted by fluids in the rocks exceeds the pressure in the borehole. They can be avoided by adjusting the pressure in the borehole each time that a new type of rock is met. To detect changes in rock strata as drilling proceeds, data need to be collected from around the drill head. There are severe technical difficulties with this, but progress is being made, and the problem of detecting discontinuities in the underlying signal of well-log data, such as that shown in Fig. 1, will soon be of practical importance.

In the next section we give a brief introduction to hidden Markov models, and the specific model that we use for the well-log data. In Section 3 we describe current particle filter methods for analysing data from such a model. These all differ solely through the resampling stage, and in Section 4 we describe a new, optimal resampling algorithm. The analysis of the well-log data, both by a new particle filter, using the optimal resampling algorithm, and two existing particle filters is shown in Section 5. Our results show that it is feasible to perform on-line analysis of well-log data by using particle filters. Furthermore, the new particle filter outperforms the existing particle filters by between one and two orders of magnitude. The paper concludes with a discussion.

The data that are analysed in the paper can be obtained from

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2. Hidden Markov models

A hidden Markov model is a hierarchy with three components: a hidden state variable I_t, a measurable state X_t and measurements Y_t. The hidden state variable is a Markov chain, with finite state space $\{1, \ldots, R\}$, and satisfies

$$P(I_t = j | I_{t-1} = i) = P_{ij},$$

for some known $R \times R$ transition matrix P. Conditional on $I_t = i$, the measurable state and the measurements satisfy

$$X_t = f_i(X_{t-1}, V_t),$$
$$Y_t = g_i(X_t, W_t),$$

for some sets of functions $f_i(\cdot, \cdot)$ and $g_i(\cdot, \cdot)$, $i = 1, \ldots, R$, and where V_t and W_t are random variables representing noise.

For concreteness, and to simplify the following discussion, we present the specific hidden Markov model that we use to model the well-log data. We assume that the hidden Markov state is two dimensional, $I_t = (S_t, O_t)$, with S_t and O_t both taking values in $\{1, 2\}$. This can be formulated as a hidden Markov model with a hidden component that can take four values. Conditional on I_t, the measurable state, the expected nuclear magnetic response, satisfies

$$X_t = \begin{cases} X_{t-1} & \text{if } S_t = 1, \\ \mu + \sigma V_t & \text{if } S_t = 2, \end{cases} \tag{1}$$

and the measurements satisfy

$$Y_t = \begin{cases} X_t + \tau_1 W_t & \text{if } O_t = 1, \\ \nu + \tau_2 W_t & \text{if } O_t = 2, \end{cases} \tag{2}$$

where $\{V_t, W_t\}_{t=1,\ldots}$ are uncorrelated, standard Gaussian random variables and μ, ν, σ, τ_1 and τ_2 are suitably chosen hyperparameters. The system equation (1) allows for jumps in the underlying signal, whereas the measurement equation (2) allows for clusters of outliers. The values of the hyperparameters that we chose, and further discussion of this and possible alternative models, are given in Section 5.

3. Particle filters

Particle filters are sequential Monte Carlo algorithms designed for on-line Bayesian inference problems. The first particle filter was the Bayesian bootstrap filter of Gordon *et al.* (1993), but earlier sequential Monte Carlo algorithms exist (West, 1992). The use of particle filters for smoothing, i.e. inference about the past, was first suggested by Kitagawa (1996). See Liu and Chen (1998) for a review and Doucet *et al.* (2001), and references therein, for further details and applications of particle filters.

For the general hidden Markov model of Section 2, a simple particle filter would be as follows. At each stage the particle filter stores a set of weighted particles. Each particle consists of a possible realization of the history of both the hidden and the measurable states. To economize on storage, each particle may consist of only a summary of these histories, such as the current value of the two states, and the time of the last changepoint. The weighted particles produce a discrete approximation to the posterior distribution of the history of the states, or the suitably chosen summary. The particles are propagated through time according to the dynamics of the states, and their weights are updated, via importance sampling, to take account of the information that is in each measurement.

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Improvements on the basic particle filter algorithm include the use of better proposal distributions in the importance sampling stage (Pitt and Shephard, 1999), the use of Markov chain Monte Carlo sampling (Gilks and Berzuini, 2001; Fearnhead, 2002a) and the use of resampling (Liu and Chen, 1995; Carpenter *et al.*, 1999) which can be important to avoid having only a small number of particles with non-negligible weight.

We shall consider hidden Markov models where, conditional on the history of the hidden state $I_1, \ldots, I_t$, the posterior distribution of the history of the measurable state $X_1, \ldots, X_t$ is tractable. The most common examples of such models are those which, conditional on the history of the hidden state, are linear–Gaussian (this includes our model for the well-log data which is specified by equations (1) and (2)). In such cases the Kalman filter can be used to calculate the parameters of the multivariate Gaussian posterior distribution of the history of the measurable state given the history of the hidden state. Thus we can use the marginalization scheme of Liu *et al.* (2001), and instead of the particles consisting of realizations of histories of both states they can consist solely of realizations of the history of the hidden state, or some suitable summary. Essentially, the posterior distribution of the histories of the states is a mixture, with the number of terms in the mixture increasing exponentially over time; each particle relates to a single term in this mixture distribution. Such an approach can lead to substantial gains in efficiency (MacEachern *et al.*, 1999).

For the well-log data, we use a four-dimensional particle $\beta = \{\alpha, \gamma^2, i, \tau\}$, where α and γ^2 are respectively the mean and variance of the posterior distribution of X_t (conditional on the specific history of the hidden state which is related to the particle), i is the current value of I_t and τ is the time of the last changepoint. At stage t, a particle filter algorithm, using marginalization, is as follows.

(a) Assume that we have a set of N particles, $\{\beta_{t-1}^{(j)}\}_{j=1}^N$, with weights $\{q_{t-1}^{(j)}\}_{j=1}^N$ at time $t-1$.
(b) Using the Kalman filter, generate a set of RN particles for time t, $\{\beta_t^{*(j)}\}_{j=1}^{RN}$. Each of the particles at time $t-1$ produces R new particles (*descendants*), one for each possible value of I_t.
(c) Calculate the weights of the RN new particles. The weight of a particle β_t^*, which is a descendant of particle $\beta_{t-1}^{(j)}$, is

$$q_t^* = q_{t-1}^{(j)} P_{il}\, L(y_t|\beta_t^*),$$

where i and l are respectively the values of I_{t-1} and I_t associated with the two particles, P is the transition matrix of the hidden Markov component and L is the likelihood.
(d) Resample from the set of RN weighted particles, to produce a set of N particles, $\{\beta_t^{(j)}\}_{j=1}^N$, with weights $\{q_t^{(j)}\}_{j=1}^N$.

At stage t the distribution of the RN particles is exactly equivalent to the posterior distribution given the latest data, based on a discrete prior specified by the weighted particles at time $t-1$. Resampling at each stage is necessary to avoid an exponentially increasing number of particles, and hence an exponentially increasing computational cost for the algorithm. A slightly more efficent implementation of this algorithm would involve initially calculating only the weights of the RN particles, performing the resampling and then calculating the actual values of the N particles that are kept.

Algorithms of this type include the detection estimation algorithm of Tugnait (1982) and the random-sampling algorithm of Akashi and Kumamoto (1977). In the detection estimation algorithm, the resampling stage is deterministic, with the N particles with largest weights being kept. In the random-sampling algorithm, resampling is stochastic, with precisely one descendant of each of the N particles at time $t-1$ being kept. For each particle, a descendant is chosen

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with probability proportional to the descendant's weight. The weight of this chosen particle is then changed to be the sum of the weights of all the parental particle's descendants.

The mixture Kalman filter is closely related to the random-sampling algorithm but has an extra resampling phase. After each iteration, the coefficient of variation of the weights of the particles is calculated. If this is greater than some predetermined value, chosen arbitrarily by the user, then an extra resampling stage occurs, in which a set of N equally weighted particles is obtained via multinomial sampling of the current particles, with the probabilities being proportional to the particles' weights. Lower variance resampling schemes (Liu and Chen, 1998; Carpenter *et al.*, 1999) are preferable to multinomial resampling.

4. Optimal sampling

The three algorithms that were mentioned in the previous section differ solely at the resampling stage. We now consider how to perform this resampling stage optimally. The problem reduces to that of approximating a discrete probability mass function with finite support by a *stochastic* probability mass function, with fewer support points. Writing $q = \{q_j\}_{j=1}^M$ for the discrete probability mass function and $Q = \{Q_j\}_{j=1}^M$ for a stochastic approximation, the objective is to determine Q so that, for a given value of $N < M$,

(a) $E(Q_j) = q_j$, $j = 1, \ldots, M$,
(b) the support of Q has no more than N points and
(c) $E\,\Sigma_{j=1}^M (Q_j - q_j)^2$ is minimized.

Note that, from condition (a), an unbiased estimator of the expected value of any random variable Y defined for the probability mass function q is obtained simply by replacing q with Q, since

$$E\left(\sum_{j=1}^M y_j Q_j\right) = \sum_{j=1}^M y_j q_j.$$

The solution to this problem can be obtained from the following theorem.

Theorem 1. Let $N < M$ and let c be the unique root of $\Sigma_{j=1}^M \min(cq_j, 1) = N$. If an approximation Q can be found, satisfying the conditions

(a) $E(Q_j) = q_j, j = 1, \ldots, M$, and
(b) the support of Q has no more than N points, with marginal distributions given by

$$Q_j = \begin{cases} q_j/p_j, & \text{with probability } p_j, \\ 0, & \text{otherwise,} \end{cases} \tag{3}$$

where $p_j = \min(cq_j, 1)$, for $j = 1, \ldots, M$, then Q minimizes the expected squared error loss, $E\,\Sigma_{j=1}^M (Q_j - q_j)^2$, under the conditions stated.

For the proof, see Appendix A.

Resampling is equivalent to assigning a new, random, weight to each of the $M = RN$ particles $\{\beta_t^{*(j)}\}_{j=1}^M$. If this weight is 0, then that particle can be removed, and so the resampled particles are purely those particles which have non-zero weight. The random variables $Q_1, \ldots, Q_M$ represent the new weights for the M particles, and $q_1, \ldots, q_M$ represent the weights of the particles before resampling.

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Condition (a) of theorem 1 ensures that the resampling stage is unbiased and is closely related to the idea of maintaining properly weighted samples (Liu and Chen, 1998). Both the random-sampling algorithm's and mixture Kalman filter's resampling stages are unbiased. Condition (b) of theorem 1 is to ensure that at most N particles are resampled.

It remains to show that a stochastic probability mass function Q, as prescribed above, exists. We illustrate the method of construction with reference to the specific problem of resampling at stage t of a particle filter, but the method is quite general.

Step 1: consider the M $(= RN)$ particles before resampling at time t. These have weights $q_t^{*(j)}$ for $j = 1, \ldots, M$. We assume that these weights sum to 1. Calculate c, the unique solution to $N = \Sigma_{j=1}^{M} \min(cq_t^{*(j)}, 1)$.
Step 2: for $j = 1, \ldots, M$, if $q_t^{*(j)} \geqslant 1/c$ then place particle j in set 1; otherwise place particle j in set 2. Assume that L particles are placed in set 1.
Step 3: use the stratified sampling of Carpenter *et al.* (1999) to resample $N - L$ particles from set 2. The expected number of times that each particle is resampled is proportional to its weight.
Step 4: the new set of N weighted particles consists of the L particles in set 1, each given its original weight, and the $N - L$ particles in set 2, each assigned a weight $1/c$.

The stratified sampling of Carpenter *et al.* (1999) which is used at step 3 is given in Appendix B. It ensures that either none or one copy of each particle is resampled. It is straightforward to see that this resampling algorithm implements the optimal solution.

At first glance the optimal resampling scheme might appear computationally intensive in comparison with the other resampling schemes. In fact, we show in Appendix C that the computational complexity is of the same order and the scheme can be implemented in an algorithm with $O(N)$ average running time. As the new particle filter differs from these existing algorithms only through the resampling procedure, it will thus have a similar computational cost.

The key feature of the resampling algorithm is that there are no multiple copies of particles in the final set of N particles. For problems where the state space of the particles is continuous, there is an advantage in having multiple copies of particles, which can probe the future independently. However, the discrete nature of this problem makes a subtle difference. Each retained particle can have multiple descendants—up to R possible distinct descendants, one for each possible value of I_t—and thus each has the potential to probe the future as much as is possible. Having multiple copies of a particle is wasteful: the same information can be contained in a single particle, which has a weight equal to the sum of the weights of the copies of the particle.

The resampling algorithm has the advantage of the detection estimation algorithm, in that a set of particles with largest weight are automatically resampled, but without the biasedness. The random-sampling algorithm also ensures that there are no multiple copies of any particle, but it suffers from the constraint that only one descendant from any particle can survive the resampling stage, even if more than one descendant has a large weight. This constraint can lead quickly to a skewedness in the weights of the particles. The above resampling algorithm has no such constraint.

There are similarities between this optimal resampling algorithm and both rejection control (Liu *et al.*, 1996) and partial rejection control (Liu *et al.*, 2001). All three resampling schemes keep particles whose weight is above some threshold and then resample from the remaining particles. In both forms of rejection control, the resampling is performed by first accepting or rejecting each particle with a probability that is proportional to its weight. Accepted particles are reweighted, and rejected particles are replaced. The two forms of rejection control differ in how rejected particles are replaced. Both algorithms were developed for more standard applications

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of particle filters, and neither approach is immediately applicable to the resampling problem that we consider here, that of resampling a set of particles to produce a smaller, fixed, number of particles.

5. Well-log data

We analysed the well-log data of Fig. 1 by using the particle filter of Section 3 and the resampling algorithm of Section 4. Our aim is

(a) to investigate whether particle filters are suitable for on-line analysis of well-log data and
(b) to see how the resampling algorithm that was introduced in Section 4 improves the performance of the particle filter.

To address aim (b), we compared three particle filter algorithms: one using the new resampling scheme, the mixture Kalman filter algorithm and, as a bench-mark, the simplest possible resampling scheme, that of multinomial sampling.

There are two sources of error when analysing the well-log data. Firstly, there is the potential error in the true posterior due to a misspecification of the system or observation models. Secondly, there are the errors in the approximation of this posterior by the particle filter. It is the second error that we can, and will, focus on. As far as the first error is concerned, we aim purely to show that the model that we use produces a posterior that is qualitatively accurate. Further refinement of the model can potentially lead to more accurate results. The extent to which the model can be refined, and yet the marginalization technique, and hence the particle filter of Section 3, can still be used is discussed below.

This section is subdivided into three. Firstly we address and discuss how to model the well-log data. Secondly we analyse how the particle filter performs at on-line detection of the changepoints. Finally we compare the new resampling algorithm of Section 4 with two alternatives.

The well-log data of Fig. 1 have been analysed previously by Ó Ruanaidh and Fitzgerald (1996), who batch processed the data by using a Gibbs sampler. Outliers were removed by hand before the data were analysed, and the number of changepoints was fixed before the analysis. By contrast, in on-line analysis by the particle filter, decisions about outliers and the number of changepoints can be dealt with automatically.

5.1. Modelling the well-log data

We based our system model, specified by expression (1), on that of Ó Ruanaidh and Fitzgerald (1996). Such a model produces the step function form for the underlying signal that is evident from the data. We use an informative prior for the height of the steps with $\mu = 115000$ and $\sigma = 10000$, and we assume a constant probability of discontinuity, $P(S_t = 2) = 1/250$, independently of $S_1, \ldots, S_{t-1}$, for all t. Since our primary concern is with identifying the times at which discontinuities occur these prior assumptions are not overly prescriptive.

Ideally the observation model would be constructed from different, but similar, data sets. These would have been used to learn about the error structure in the observation process. However, since such data sets were not available, hyperparameters for the observation model were obtained by a visual inspection of the data.

Some outliers below the main body of data are apparent. This motivated the model that we have used, expression (2). When the Markov chain O_t is in state 1, the observations will be modelled as the true state corrupted by additive noise. State 2 will represent an outlier state, and, for simplicity, the observation will be modelled as a draw from a Gaussian random variable whose parameters are independent of the true state. This can be seen as attempting to mimic the

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approach of Ó Ruanaidh and Fitzgerald (1996). They removed the outliers, which is equivalent to ignoring any information about the state of the process that they may contain.

Around 70 observations appear to be outliers. These occur in 16 clusters. This suggests that suitable values of the transition probabilities would be approximately $P(O_t = 2 | O_{t-1} = 1) = 0.004$ and $P(O_t = 2 | O_{t-1} = 2) = 0.75$. The outlier distribution was taken to have a mean ν of 85000 and a standard deviation τ_2 of 12500. The standard deviation τ_1 of non-outlying observations was taken to be 2500.

Ó Ruanaidh and Fitzgerald (1996) assumed additive Laplacian noise for the data. The analysis of the well-log data by a particle filter under such a model can be found in Fearnhead (1998). More complicated models, which include more detailed modelling of the outliers, and which allow for correlated noise, were also considered there. For all these models the posterior distribution of the history of the state X_t, conditional on the history of the hidden state I_t, could be calculated analytically (Fearnhead, 1998). Hence, the well-log data could still be analysed by using the particle filter of Section 3.

5.2. Analysis by new particle filter

We now analyse the well-log data, using the model specified by expressions (1) and (2), and hyperparameters defined as above. Here, we consider solely the results based on the particle filter of Section 3, and the resampling algorithm of Section 4. Under our model, distinct particles at time t can become identical when propagated to time $t + 1$, because the new value of the state after a changepoint is independent of the old value. In implementing the particle filter, we merged all particles which were identical, i.e. all particles with $S_t = 2$ and the same value for O_t, before resampling. This merging of particles is not possible for the resampling algorithm of the mixture Kalman filter.

The main aim of analysing the well-log data is to detect the changepoints in the signal on line. So, after processing each measurement, the probability of a jump having occurred during the last k time points was estimated. The results that we present are for $k = 5$, but similar results were obtained with slightly different values of k.

The results of a single on-line analysis of the well-log data by the new particle filter are shown in Fig. 2. The particle filter used 100 particles and analysed the complete data in 1 min. Fig. 2 shows both the estimates of a recent changepoint and the estimates that the current measurement is an outlier. Differentiation between a changepoint and a cluster of outliers is particularly difficult for on-line analyses. Note that the estimates that are plotted at time t are for the posterior probabilities based only on observations up to time t.

The filter appears to have performed well, with all obvious changepoints having posterior probabilities close to 1. In a few cases, the filter appears to have misclassified outliers as changepoints. An easier evaluation of the performance of the filter can be gained from looking at an estimate of the underlying signal (Fig. 1).

The estimate was obtained from the output of the particle filter. The changepoints were fixed to be at times where the posterior probability of a recent jump was greater that 0.9, of which there were 16. The choice of the value 0.9 to determine jumps is arbitrary, but similar results are observed for different choices of this cut-off value. The value of the state between each pair of adjacent changepoints was estimated by the mean of all measurements in that time period which had negligible probability of being outliers.

5.3. Comparison of particle filters

We compared the performance of three particle filters at analysing the well-log data: the new particle filter, the mixture Kalman filter and a particle filter using multinomial resampling,

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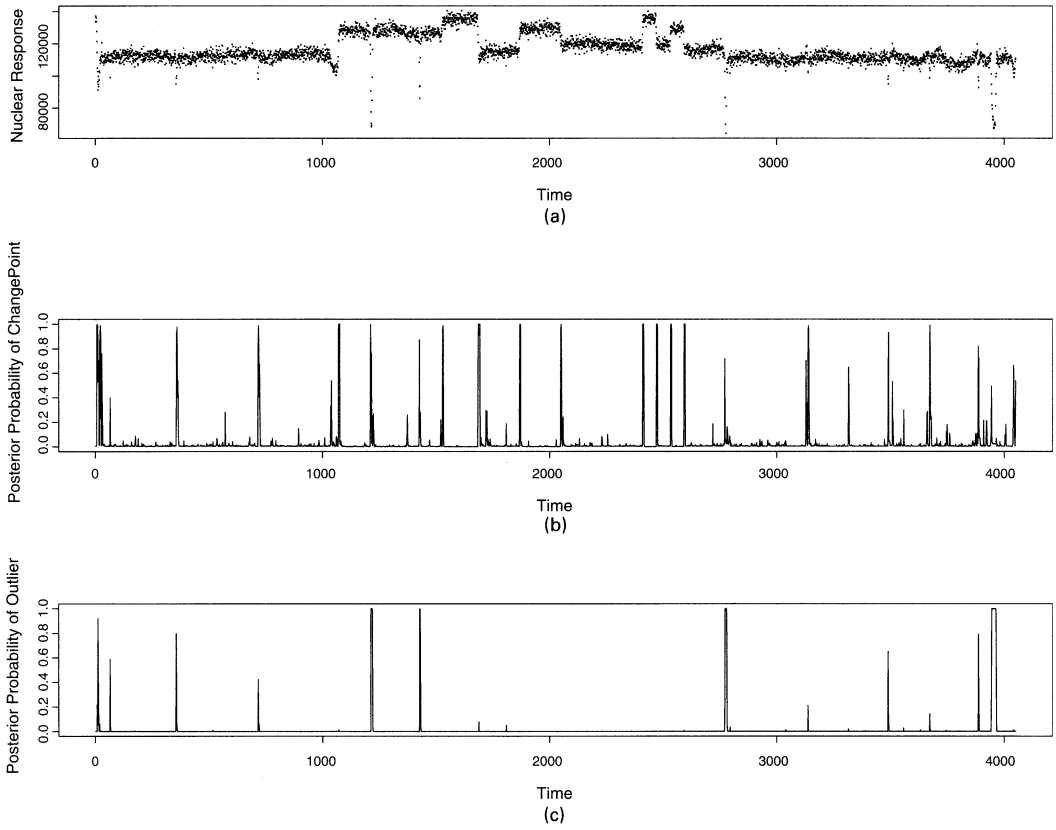

Fig. 2. Results of the on-line analysis of the well-log data by the new particle filter, using 100 particles: (a) the data, (b) estimates of the probabilities of a recent changepoint and (c) probability that a measurement is an outlier

which we call the *basic particle filter*. Of primary concern is the comparison of the new particle filter with the mixture Kalman filter. The results for the basic particle filter are included to demonstrate how important the resampling stage of the particle filter algorithm is. The computational cost of each filter is very similar.

Our comparison is based on the accuracy with which each filter approximates the true posterior density. To perform this comparison we obtained an accurate approximation of this posterior by running the new particle filter with 100 000 particles. This took approximately 2 h computing time.

Each of the three particle filters analysed the data 100 independent times. For each run of a particle filter, 100 particles were used. In the mixture Kalman filter, extra resampling occurred when the coefficient of variation of the weights became greater than 10. Similar results for the mixture Kalman filter were observed for coefficients of variation of 2, 3 and 5. On average, between 10 and 20 such resampling stages occurred during each run of the mixture Kalman filter.

Firstly we measured the performance of each particle filter on the basis of its accuracy at estimating the posterior probability of a recent changepoint at each time point. The results for both the absolute and the mean-square error of these estimates is given in Table 1. Secondly we took a decision theoretic approach. For each run of each particle filter, we inferred a

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Table 1. Comparison of performance of three particle filter algorithms (see the text for details) at analysing the well-log data†

Algorithm	*Absolute error*	*Square error*
New particle filter	3.66	0.49
Mixture Kalman filter	73.6	19.0
Basic particle filter	143	88.7

†Each particle filter used 100 particles. At each time point, each filter estimates the probability that a changepoint occurs during the five previous time points. This estimate is compared with an accurate estimate of the probability, based on output from a particle filter with 100 000 particles. The absolute error (or square error) is the average, over 100 runs of each filter, of the sum of the absolute (or squared) differences between that filter's estimates, and the accurate estimates, over the 4050 time points.

Table 2. Comparison of performance of three particle filter algorithms (see the text for details) at analysing the well-log data†

Algorithm	*False positive results*	*Missed changepoints*
New particle filter	0	0
Mixture Kalman filter	4.5	1.1
Basic particle filter	25.5	0.03

†Each particle filter used 100 particles. The results show the average number of non-existent changepoints that each algorithm inferred, and the average number of true changepoints that each algorithm missed, over the 100 runs of each filter. The true changepoints are based on an accurate estimate of the posterior from a particle filter with 100 000 particles.

changepoint whenever the probability of a recent changepoint was greater than 0.9. Often there is a cluster of consecutive times where this posterior probability is greater than 0.9; we inferred a single changepoint for each such cluster. We compared these inferred changepoints with the 16 changepoints inferred from the accurate approximation to the posterior. We characterized each particle filter by its rate of false positive results and its rate of missed changepoints. The results are in Table 2.

The performance of the new particle filter is particularly impressive. On each of the 100 runs it inferred precisely the same 16 changepoints that were inferred from the accurate approximation of the posterior. By comparison, the basic particle filter inferred a substantial number of false changepoints, and the mixture Kalman filter did not infer some of the true changepoints. In terms of the absolute, and squared, error loss functions, the new particle filter outperformed the two alternative particle filters by between one and two orders of magnitude.

6. Discussion

We have shown that particle filters are suitable tools for the on-line analysis of well-log data. Despite using a simple model, the particle filter could give qualitatively good results for detecting changepoints in the underlying signal. We have also introduced a new particle filter, based on the

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idea of an optimal resampling algorithm, and the algorithm was shown to be computationally efficient. This new particle filter outperformed two alternative filters by between one and two orders of magnitude.

One reason for the improvement in performance of the new particle filter over the mixture Kalman filter is that merging identical particles can be performed in the new particle filter but is not possible with the mixture Kalman filter resampling algorithm. However Fearnhead (2002b) has demonstrated a similar improvement in performance of the new particle filter over one based on the mixture Kalman filter for the problem of estimating parameters from mixture models, where merging identical particles is impossible for either algorithm. Thus some improvement in efficiency for the new particle filter over the mixture Kalman filter is likely in some generality.

A comparison of the new resampling algorithm with that of the mixture Kalman filter gives some insight into when the new particle filter will outperform the mixture Kalman filter. If, at resampling, the weights of the RN potential particles at the next time point are very even, then the new resampling algorithm is very similar to just resampling using the stratified sampling algorithm of Carpenter *et al.* (1999). In this case little, if any, improvement over the mixture Kalman filter is likely. By comparison, if the weights are skewed the new resampling algorithm can be extremely efficient, and the new particle filter can substantially outperform the mixture Kalman filter. In the extreme case where fewer than N of the RN potential particles at the next time point have non-negligible weight, one step of the new particle filter will be, essentially, exact, i.e. the discrete distribution that is specified by the N resampled particles, and their weights, will be almost identical to the discrete distribution of the RN weighted particles before resampling. The same would be true of the mixture Kalman filter only if the particles with non-negligible weight are descended from distinct particles at the current time point.

One advantage that the mixture Kalman filter has over the new particle filter is in the storage of memory. The mixture Kalman filter resampling can be done in stages, with the descendant of each current particle being sampled in turn. The new particle filter requires that the weights of all the RN possible particles at the next time point be calculated before resampling. Thus the memory requirement of the resampling is increased from $N + R - 1$ to RN. This is likely to be a problem only for very large R, when any method which is based on simulating the value of the hidden state at time t from its posterior distribution, which requires a calculation of $O(R)$ for each particle, is likely to be impracticable.

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Appendix A: Proof of theorem 1

The condition that, with probability 1, at most N of the Q_js are non-zero implies that marginally Q_j is equal to some random variable C_j with probability p_j and is 0 otherwise. Further, it implies that $\Sigma_{j=1}^{M} p_j \leqslant N$, and since

$$E\{(Q_j - q_j)^2\} = p_j\{E(C_j) - q_j\}^2 + p_j \operatorname{var}(C_j) + (1 - p_j)q_j^2$$

the expected squared error loss is minimized by the choice $C_j = c_j$, a constant, for each j. Further, the unbiasedness condition $E(Q_j) = q_j$ implies that $c_j = q_j/p_j$.

With this choice of Q, we have

$$E\left\{\sum_{j=1}^{M}(Q_j - q_j)^2\right\} = \sum_{j=1}^{M} q_j^2(1/p_j - 1).$$

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Minimizing this is equivalent to minimizing $\Sigma_{j=1}^{M} q_j^2/p_j$, subject to $\Sigma_{j=1}^{M} p_j \leqslant N$. This is achieved by expression (3).

Appendix B: Stratified resampling

Stratified sampling, as required for the new particle filter, can be implemented as follows. Let w_i, $i = 1, \ldots, M - L$, be the weights of the particles to be resampled in set 2. A total of $N - L$ particles need to be resampled. This algorithm is a version of that given in Carpenter *et al.* (1999).

Initialize: let $K = \Sigma_{i=1}^{M-L} w_i/(N - L)$ and let U be the realization of a uniform random variable on $[0, K]$. Set $i = 1$.
While $i \leqslant (M - L)$
  set $U = U - w_i$.
  If $U < 0$
    resample particle i;
    set $U = U + K$.
  Set $i = i + 1$.

Appendix C: Computational complexity of optimal resampling

Theorem 2. The optimum resampling scheme can be calculated with an expected running time that is linear in the number of particles, i.e. $O(N)$.

Proof. We prove that the resampling is $O(M) = O(RN)$, from which it follows, because R is constant, that the resampling is $O(N)$. First note that, if κ is the smallest element of the set $q = \{q_1, \ldots, q_M\}$ satisfying

$$\sum_{j=1}^{M} \min\left(\frac{q_j}{\kappa}, 1\right) \leqslant N, \tag{4}$$

then $c = (N - A_\kappa)/B_\kappa$, where A_κ is the number of elements of q that are larger than κ and B_κ is the sum of the remaining elements. Inequality (4) can be rewritten as

$$\kappa^{-1} B_\kappa + A_\kappa \leqslant N. \tag{5}$$

The value κ can be found by a selection algorithm. The method is similar to that of Blum *et al.* (1973), who gave the original $O(M)$ algorithm for finding the median of M numbers. The method is recursive and organized as follows:

(a) `Random-partition`, i.e. segregate the weights about a trial value of κ;
(b) check adjusted criterion (5);
(c) recurse on the appropriate partition.

For a list of M elements, `Random-partition` has average running time $O(M)$ (Cormen *et al.*, 1996). As a by-product, without increasing the order of computational complexity, the number in each partition and the sum of the values in the partition can be obtained, either directly or by subtraction. These quantities are used to adjust criterion (5) appropriately in the respective partitions. Writing $T(M)$ for the average time complexity of finding κ in a partition of size M, we have

$$\begin{aligned} T(M) &\leqslant \frac{1}{M}\left\{T(M-1) + 2\sum_{k=M/2}^{M-1} T(k)\right\} + O(M) \\ &\leqslant \frac{2}{M}\sum_{k=M/2}^{M-1} T(k) + O(M) \end{aligned} \tag{6}$$

where the $O(M)$ term is the running time of `Random-partition`, including the additional recording of the totals and sizes of the partitions. Making the inductive hypothesis that $T(k) \leqslant Ck$ for $k < M$, inequality (6) becomes, after some reduction,

$$T(M) \leqslant C(\tfrac{3}{4}M - \tfrac{1}{2}) + O(M) \leqslant CM,$$

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where C is chosen so that $C(M/4+\frac{1}{2})$ dominates the $O(M)$ term. For greater detail on this type of algorithmic analysis see Cormen *et al.* (1996), page 189. Once κ has been found the remainder of the calculations for the resampling scheme are immediate, since `Random-partition` will have already identified those particles which are to be retained without adjustment and the $N-L$ remaining particles are sampled in linear time by using the algorithm of Carpenter *et al.* (1999).

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